

Marketing Campaign Dashboard

Documentation & Design Review Report

Report scope	4 dashboard pages (Power BI)
Pages covered	Overview Customer Demographics Spending by Product Category Purchase Channels & Campaigns
Dataset	marketing_campaign_cleaned.csv (2,236 customers)

Executive Summary

This dashboard was built iteratively across four pages, each covering a distinct angle of the marketing campaign dataset: a high-level Overview, Customer Demographics, Spending by Product Category, and Purchase Channels & Campaign Performance. Development followed a consistent principle throughout: every new chart had to add information that no other chart on the page already showed, rather than repeating the same numbers in a different visual shape. Where a chart duplicated an existing one (e.g. a Bar Chart mirroring a Donut Chart, or a Waterfall re-stating a simple total), it was identified and replaced with a genuinely complementary view (such as average instead of sum, or a customer-level relationship instead of a category-level total).

Build Approach

Each page was scoped first (which questions should it answer), then built one visual at a time. Two recurring technical patterns came up repeatedly during the build and are worth calling out since they explain several of the design decisions documented in this report:

- **Unpivot for multi-column comparisons:** Whenever several columns needed to be compared side by side as categories (e.g. the six spending columns, the four channel columns, the five campaign columns), the columns were reshaped in Power Query using Unpivot into a single Category/Value pair. This was done on a duplicated copy of the table each time, so the original wide-format table stayed intact for visuals that still needed it (like the Scatter Charts).
- **Row-level detail for Scatter Charts:** By default, Power BI's Scatter Chart aggregates data to the level of whatever field is in Legend, producing only a handful of points. Adding the customer's ID field to the Values/Details bucket forces one point per customer instead, which was necessary for both Scatter Charts in this dashboard (Income vs Spending, and Web Visits vs Web Purchases).

Page 1: Overview

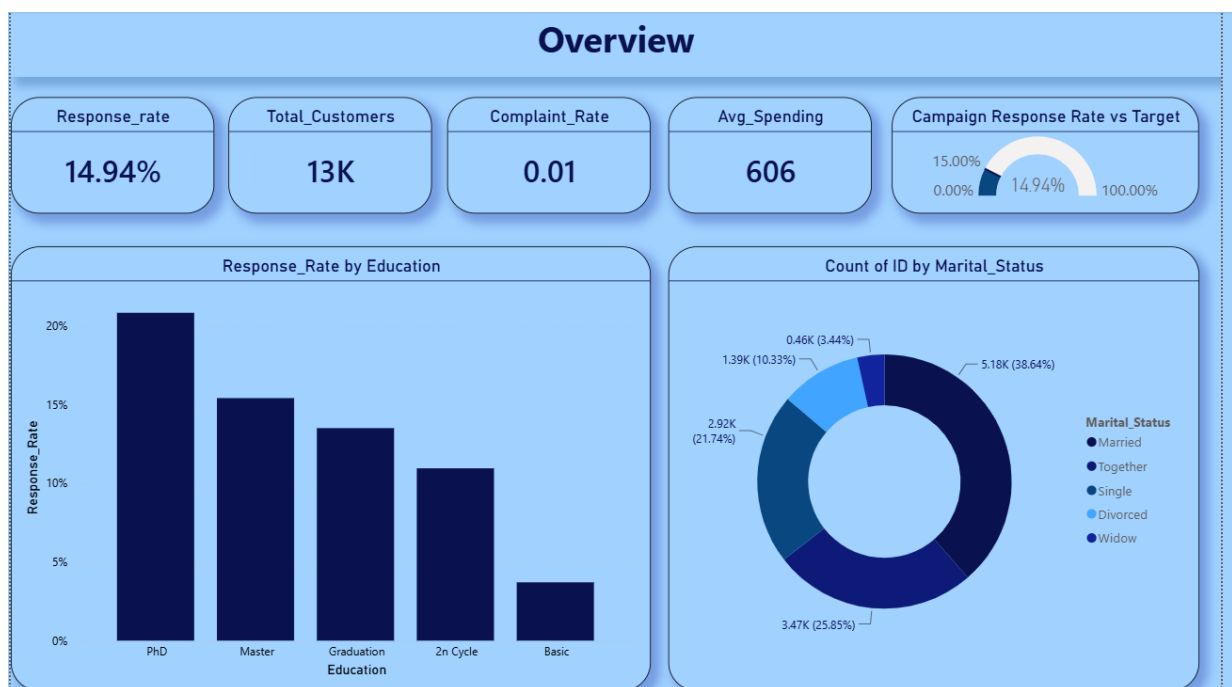
Purpose: A landing page giving a single-glance read of overall campaign health before drilling into detail pages.

Visuals included

- 4 KPI Cards: Response Rate, Total Customers, Complaint Rate, Avg Spending
- Gauge: Campaign Response Rate vs Target
- Column Chart: Response Rate by Education
- Donut Chart: Count of Customers by Marital Status

Key insights the page surfaces

- Overall response rate sits at 14.94% against a stated target
- Response rate is highest among PhD-educated customers and drops sharply for the Basic education group
- Married and Together customers make up roughly two-thirds of the customer base



Screenshot — Overview page as built in Power BI

Page 2: Customer Demographics

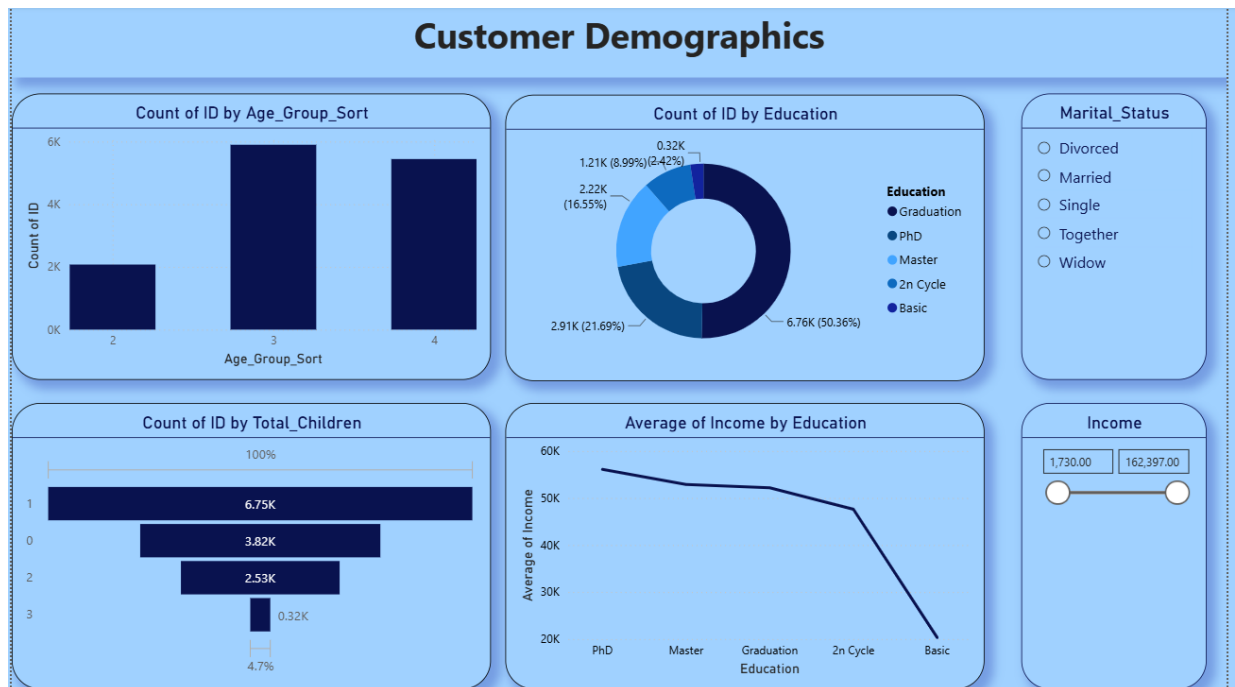
Purpose: Profiles who the customers are, independent of spending or campaign behavior, to support audience segmentation.

Visuals included

- Column Chart: Customer Count by Age Group
- Donut Chart: Customer Count by Education
- Slicer: Marital Status
- Funnel-style Bar Chart: Customer Count by Total Children
- Line Chart: Average Income by Education
- Slicer: Income (range)

Key insights the page surfaces

- Graduation-level customers are the largest education segment (over 50%)
- Households with 1 child are the most common customer group
- Average income declines steadily from PhD down to Basic education, with a steep drop at the Basic tier



Screenshot — Customer Demographics page as built in Power BI

Page 3: Spending by Product Category

Purpose: Answers 'what do customers buy' by breaking total spend down across product categories and linking it to income.

Visuals included

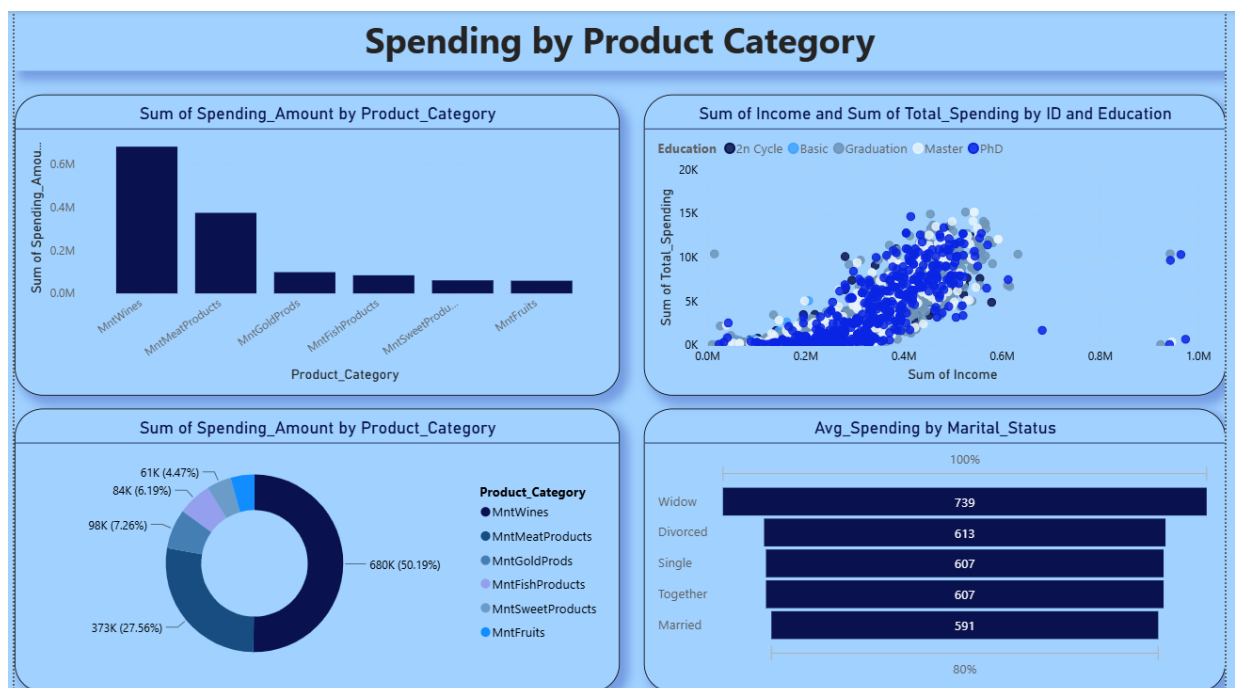
- Column Chart: Total Spending by Product Category (absolute values)
- Donut Chart: Share of Total Spending by Product Category (percentages)
- Scatter Chart: Income vs Total Spending, one point per customer, colored by Education
- Bar Chart: Average Spending by Marital Status

Key insights the page surfaces

- Wine dominates spending, accounting for roughly half of total product spend, followed by Meat products
- There is a clear positive relationship between income and total spending
- Widowed customers show the highest average spend per person despite being a small segment

Design notes / decisions made during build

- The Column and Donut Chart intentionally show the same six categories but in complementary forms: absolute totals vs. percentage share, which is a deliberate part of the design pattern used across the whole dashboard.
- The Scatter Chart required an Unpivot step and, separately, adding Customer ID to the Values field — without ID the chart silently aggregated to only 5 points (one per education level) instead of one point per customer.



Screenshot — Spending by Product Category page as built in Power BI

Page 4: Purchase Channels & Campaign Performance

Purpose: Combines how customers purchase (channel mix) with how they respond to marketing (campaign acceptance), originally scoped as two separate pages and merged during build.

Visuals included

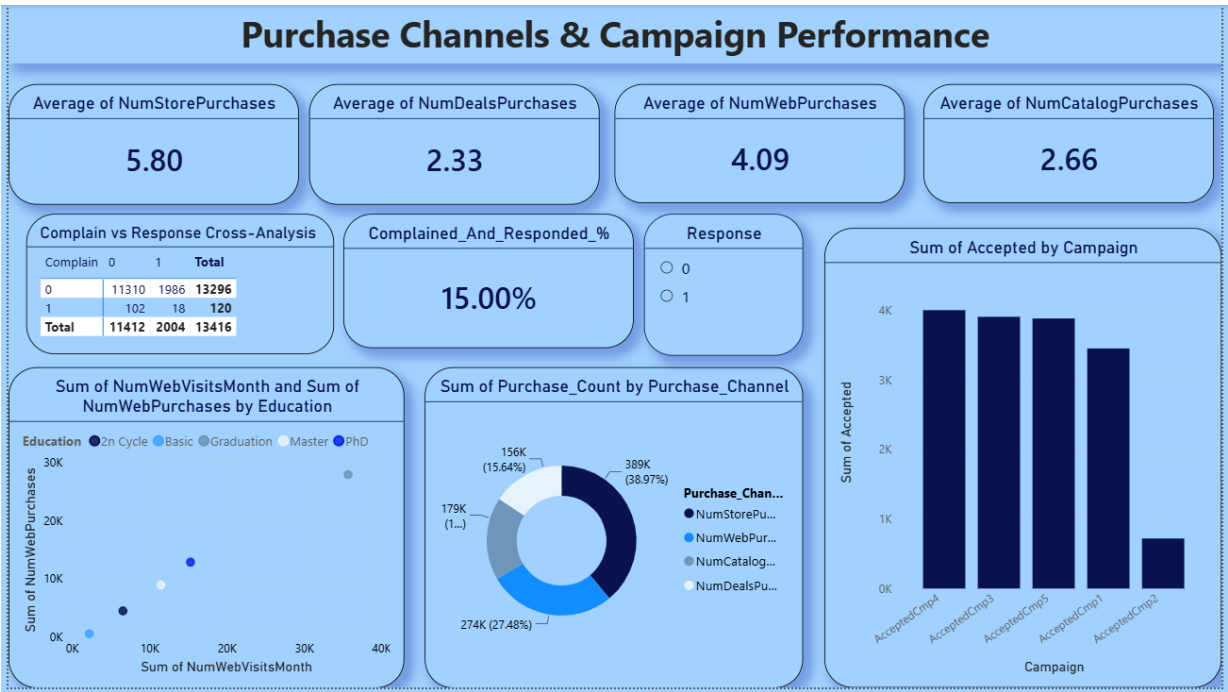
- 4 KPI Cards: Average purchases per channel (Store, Web, Catalog, Deals)
- Matrix: Complain vs Response cross-tabulation
- Card: % of complaining customers who also responded to the current campaign
- Slicer: Response (current campaign)
- Scatter Chart: Web Visits vs Web Purchases per customer, colored by Education
- Donut Chart: Share of Purchases by Channel
- Bar Chart: Acceptance count by past Campaign (Cmp1–Cmp5)

Key insights the page surfaces

- Store purchases have the highest per-customer average, ahead of Web, Catalog, and Deals
- Campaign 4 had the strongest historical acceptance, while Campaign 2 performed far below the rest
- Of the customers who filed a complaint, 15% still went on to respond positively to the current campaign, suggesting complaints don't automatically translate into disengagement

Design notes / decisions made during build

- A Bar Chart and a Waterfall Chart that both restated the same channel totals already shown in the Donut Chart were removed and replaced with the 4 average-purchase KPI Cards.
- This page combines two originally separate planned pages (Purchase Channels and Campaign Performance) into one, resulting in 7 visual elements versus 4 on each of the other pages.



Screenshot — Purchase Channels & Campaign Performance page as built in Power BI

Recommendations for Next Iteration

- Consider splitting Page 4 back into two pages (Purchase Channels / Campaign Performance) if it feels crowded on smaller screens.
- Add a shared slicer (e.g. Date range or Marital Status) that filters across all four pages for cross-page interactivity.
- Rename default visual titles (e.g. 'Sum of Purchase_Count by Purchase_Channel') into short, human-readable titles for a more polished, presentation-ready feel.
- Add tooltips or a short legend note on the Complain-vs-Response Matrix given the very small sample size (120 complaining customers) so readers don't over-interpret the 15% figure.